Algebra I: Midsemester Practice Paper

One-hour, Quiz-1-style mock • 20 marks

Instructions. Allow exactly 60 minutes. Attempt *one* option from each of Questions 1–4. Question 5 is compulsory. Write one reason or counterexample for every True/False answer. No notes during the attempt. The complete solutions begin after a page break; do not look at them until the hour is over.

Suggested timing. Questions 1 and 2: 8 minutes each; Questions 3 and 4: 12 minutes each; Question 5: 15 minutes; checking: 5 minutes.

Question 1 (3 marks): choose (a) or (b)

(a) Let $f : X \to Y$, $A \subseteq X$, and $B \subseteq Y$. Define

$$f_*(A) = \{f(x) : x \in A\}, \qquad f^*(B) = \{x \in X : f(x) \in B\}.$$

Prove that $f_*(A) \subseteq B$ if and only if $A \subseteq f^*(B)$. Deduce that $A \subseteq f^* f_*(A)$.

OR

(b) Let $X = \{1, 2, 3\}$, $Y = \{a, b, c\}$, and let $f : X \to Y$ satisfy $f(1) = f(2) = a$, $f(3) = b$. For $A = \{1, 3\}$, calculate $f_*(A)$ and

$$f_!(A) = \{y \in Y : f^{-1}(\{y\}) \subseteq A\}.$$

Explain why c belongs to $f_!(A)$ even though $c \notin f(X)$.

Question 2 (3 marks): choose (a) or (b)

(a) Let G, H be groups and let $\phi : G \to H$ satisfy $\phi(xy) = \phi(x)\phi(y)$ for every $x, y \in G$. Prove that $\phi(e_G) = e_H$ and $\phi(x^{-1}) = \phi(x)^{-1}$.

OR

(b) On $S = \mathbb{R} \setminus \{-1\}$ define $a * b = a + b + ab$. Prove that $(S, *)$ is a group. State its identity and the inverse of a.

Question 3 (4 marks): choose (a) or (b)

(a) Consider

$$\phi : \mathbb{Z}/12\mathbb{Z} \longrightarrow \mathbb{Z}/8\mathbb{Z}, \qquad \phi([k]_{12}) = [2k]_8.$$

Prove that ϕ is well-defined and is a group homomorphism. Find its kernel and image.

OR

(b) List every element of $\langle [7] \rangle$ in $\mathbb{Z}/18\mathbb{Z}$ and of $\langle [5] \rangle$ in $\mathbb{Z}/20\mathbb{Z}$. Give a short reason for the number of elements in each subgroup.

Question 4 (4 marks): choose (a) or (b)

(a) Let

$$P = \langle x, y, z \mid yxyz^{-2} = e \rangle.$$

Show that P is isomorphic to the free group on y, z. State how the generator x is expressed using y, z.

OR

(b) Let G be a group and X a nonempty set. On the functions $G^X = \{f : X \to G\}$ define $(fg)(x) = f(x)g(x)$. Prove that G^X is a group by giving its identity and the inverse of an arbitrary f, and checking associativity.

Question 5 (compulsory; 6 marks)

Mark each statement *True* or *False*. Give one sentence of justification or one counterexample.

(a) For every $f : X \to Y$ and $B, C \subseteq Y$, $f^*(B \cup C) = f^*(B) \cup f^*(C)$.

(b) For every $f : X \to Y$ and $A, D \subseteq X$, $f_*(A \cap D) = f_*(A) \cap f_*(D)$.

(c) If $y \notin f(X)$, then $y \in f_!(A)$ for every $A \subseteq X$.

(d) $\langle [3] \rangle = \mathbb{Z}/16\mathbb{Z}$ under addition modulo 16.

(e) There is a nontrivial group homomorphism $\mathbb{Z}/7\mathbb{Z} \to \mathbb{Z}/15\mathbb{Z}$.

(f) If $g^2 = e$ for every element g of a group G, then G is abelian.

Worked solutions and marking guide

Read this only after completing the timed attempt.

Question 1(a)

Suppose $f_*(A) \subseteq B$. Take any $x \in A$. Then $f(x) \in f_*(A) \subseteq B$, so $x \in f^*(B)$. Hence $A \subseteq f^*(B)$.
Conversely, suppose $A \subseteq f^*(B)$. Take any $y \in f_*(A)$. By definition $y = f(x)$ for some $x \in A$. Since $x \in f^*(B)$, we have $f(x) = y \in B$. Hence $f_*(A) \subseteq B$.
Now take $B = f_*(A)$. The inclusion $f_*(A) \subseteq f_*(A)$ is automatic, so the equivalence just proved gives $A \subseteq f^* f_*(A)$.

Question 1(b)

$A = \{1, 3\}$, so $f_*(A) = \{f(1), f(3)\} = \{a, b\}$. The fibres are

$$f^{-1}(\{a\}) = \{1, 2\}, \quad f^{-1}(\{b\}) = \{3\}, \quad f^{-1}(\{c\}) = \varnothing.$$

The first fibre is not contained in A because $2 \notin A$. The second fibre is contained in A, and the empty fibre is contained in every set. Thus $f_!(A) = \{b, c\}$. In particular $c \in f_!(A)$ because its fibre is empty, so the condition "every preimage of c lies in A" is true vacuously.

Question 2(a)

Let $h = \phi(e_G)$. Since $e_G e_G = e_G$ and ϕ preserves products,

$$h = \phi(e_G) = \phi(e_G e_G) = \phi(e_G)\phi(e_G) = h^2.$$

Multiply by h^{-1} in H to obtain $e_H = h$, so $\phi(e_G) = e_H$.
For any $x \in G$,

$$\phi(x)\phi(x^{-1}) = \phi(xx^{-1}) = \phi(e_G) = e_H.$$

Similarly $\phi(x^{-1})\phi(x) = e_H$. Thus $\phi(x^{-1})$ is the two-sided inverse of $\phi(x)$, and uniqueness of inverses gives $\phi(x^{-1}) = \phi(x)^{-1}$.

Question 2(b)

Observe the useful identity

$$1 + (a * b) = 1 + a + b + ab = (1 + a)(1 + b).$$

Since $a, b \neq -1$, both factors on the right are nonzero; hence $1 + (a * b) \neq 0$ and $a * b \neq -1$. This proves closure in S.
The map $T : S \to \mathbb{R} \setminus \{0\}$, $T(a) = 1 + a$, is bijective and satisfies $T(a * b) = T(a)T(b)$. Ordinary multiplication of nonzero reals is associative, so $*$ is associative. The identity is 0 because $a * 0 = 0 * a = a$. To find an inverse b, solve

$$0 = a * b = a + b + ab \quad \Longleftrightarrow \quad b(1 + a) = -a \quad \Longleftrightarrow \quad b = -\frac{a}{1 + a}.$$

This lies in S, since $1 + b = 1/(1 + a) \neq 0$. The operation is commutative, so the same b is a two-sided inverse. Thus $(S, *)$ is a group.

Question 3(a)

If $[k]_{12} = [\ell]_{12}$, then $12 \mid (k - \ell)$. Therefore $8 \mid 2(k - \ell)$, because $2(k - \ell)$ is a multiple of 24. Hence $[2k]_8 = [2\ell]_8$, proving well-definedness.
For any $[k]_{12}, [\ell]_{12}$,

$$\phi([k]_{12} + [\ell]_{12}) = [2(k + \ell)]_8 = [2k]_8 + [2\ell]_8 = \phi([k]_{12}) + \phi([\ell]_{12}).$$

Thus ϕ is a homomorphism. Its kernel consists of $[k]_{12}$ for which $[2k]_8 = [0]_8$, meaning $8 \mid 2k$, or $4 \mid k$. Hence

$$\ker \phi = \{[0]_{12}, [4]_{12}, [8]_{12}\}.$$

The outputs are exactly the even residue classes modulo 8:

$$\operatorname{im} \phi = \{[0]_8, [2]_8, [4]_8, [6]_8\}.$$

Question 3(b)

In an additive group, $\langle [d] \rangle = \{[kd] : k \in \mathbb{Z}\}$. Since $\gcd(7, 18) = 1$, the least positive k with $18 \mid 7k$ is 18. Thus [7] has order 18 and generates all of $\mathbb{Z}/18\mathbb{Z}$:

$$\langle [7] \rangle = \{[0], [1], \ldots, [17]\}.$$

Since $\gcd(5, 20) = 5$, the least positive k with $20 \mid 5k$ is 4. The first four multiples are distinct, so

$$\langle [5] \rangle = \{[0], [5], [10], [15]\} \subseteq \mathbb{Z}/20\mathbb{Z}.$$

This uses the formula $\operatorname{ord}([d]_n) = n/\gcd(n, d)$.

Question 4(a)

The relation $yxyz^{-2} = e$ is equivalent to $yxy = z^2$, hence

$$x = y^{-1}z^2y^{-1}.$$

Thus y, z generate P. Let $F(y, z)$ be the free group on two symbols y, z. The assignment sending its generators to the corresponding elements of P extends to a surjective homomorphism $\alpha : F(y, z) \to P$.
In the other direction, assign the generators of P the values

$$y \mapsto y, \qquad z \mapsto z, \qquad x \mapsto y^{-1}z^2y^{-1} \quad \text{in } F(y, z).$$

The relator maps to $y(y^{-1}z^2y^{-1})yz^{-2} = e$, so this assignment defines a homomorphism $\beta : P \to F(y, z)$. The composite $\beta\alpha$ fixes y, z, so it is the identity on $F(y, z)$. The composite $\alpha\beta$ fixes y, z and also fixes x by the displayed relation, so it is the identity on P. Therefore $P \cong F(y, z)$.

Question 4(b)

For $f, g \in G^X$, $f(x)g(x) \in G$ for every x, so the pointwise product is again a function $X \to G$ (closure). For $f, g, h \in G^X$ and arbitrary $x \in X$,

$$((fg)h)(x) = (f(x)g(x))h(x) = f(x)(g(x)h(x)) = (f(gh))(x),$$

using associativity in G. Equality at every x gives associativity of functions.

The identity is the constant function $E(x) = e_G$ because $(Ef)(x) = e_G f(x) = f(x) = (fE)(x)$. For $f \in G^X$, define $f^{-1}(x) = f(x)^{-1}$. Then $(ff^{-1})(x) = e_G = (f^{-1}f)(x)$ for all x, so f^{-1} is a two-sided inverse. Hence G^X is a group.

Question 5

(a) **True.** $x \in f^*(B \cup C)$ exactly when $f(x) \in B$ or $f(x) \in C$.

(b) **False.** Let $X = \{1, 2\}$, let f be constant with value a, and take $A = \{1\}$, $D = \{2\}$. Then $f_*(A \cap D) = \varnothing$, while $f_*(A) \cap f_*(D) = \{a\}$.

(c) **True.** If $y \notin f(X)$, its fibre $f^{-1}(\{y\})$ is empty; $\varnothing \subseteq A$ for every A.

(d) **True.** $\gcd(3, 16) = 1$, so $[3]$ has order 16 and generates all of $\mathbb{Z}/16\mathbb{Z}$.

(e) **False.** The image of $[1]_7$ must have order dividing 7. Since $\mathbb{Z}/15\mathbb{Z}$ has no nonzero element of order 7, it must map to $[0]$, making the homomorphism trivial.

(f) **True.** $g^2 = e$ implies $g^{-1} = g$ for every g. Thus $ab = (ab)^{-1} = b^{-1}a^{-1} = ba$.